

Jacob (Jake) Carlson

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<https://jscarlson.github.io>

EDUCATION

Harvard University, Cambridge, Massachusetts
Ph.D. Economics, August 2022 – May 2027 (expected)

Yale University, New Haven, Connecticut
B.A. Statistics and Data Science, August 2016 – May 2020
B.A. Ethics, Politics, and Economics, August 2016 – May 2020
Summa cum laude with distinction in both subjects

HONORS AND AWARDS

- PhD Fellowship Finalist, Two Sigma
- Presidential Scholar, Harvard University
- Phi Beta Kappa, Yale University
- Undergraduate Thesis Prize Winner, Forethought Foundation for Global Priorities Research

WORKING PAPERS

- Carlson, Jacob and Neil Shephard. “When are time series predictions causal? The potential system and dynamic causal effects.” Working paper (2026).
- Carlson, Jacob. “Making Interpretable Discoveries from Unstructured Data: A High-Dimensional Multiple Hypothesis Testing Approach.” Working paper (2025).
- Carlson, Jacob and Melissa Dell. “A Unifying Framework for Robust and Efficient Inference with Unstructured Data.” Working paper (2025).
- Andrews, Isaiah, Harvey Barnhard, and Jacob Carlson. “True and Pseudo-True Parameters.” Working paper (2024). Revision requested, *Econometric Theory*.

PUBLICATIONS

- Carlson, Jacob, Tom Bryan, and Melissa Dell. “Efficient OCR for Building a Diverse Digital History.” *ACL* (2024). **Oral presentation**.
- Dell, Melissa, Jacob Carlson, Tom Bryan, Emily Silcock, Abhishek Arora, Zejiang Shen, Luca D’Amico-Wong, Quan Le, Pablo Querubin, Leander Heldring. “American Stories: A Large-Scale Structured Text Dataset of Historical U.S. Newspapers.” *NeurIPS Datasets and Benchmarks Track* (2023).
- Bryan, Tom, Jacob Carlson, Abhishek Arora, Melissa Dell. “EfficientOCR: An Extensible, Open-Source Package for Efficiently Digitizing World Knowledge.” *EMNLP System Demonstrations* (2023).
- Carlson, Jacob, Trevor Incerti, and P.M. Aronow. “Dyadic Clustering in International Relations.” *Political Analysis* (2024).
- Shen, Zejiang, Ruochen Zhang, Melissa Dell, Benjamin Lee, Jacob Carlson, and Weining Li. “LayoutParser: A Unified Toolkit for

Deep Learning Based Document Image Analysis.” *ICDAR* (2021). **Oral presentation.**

EMPLOYMENT	Harvard University , Cambridge, Massachusetts Data Science Fellow in Deep Learning September 2020 – May 2022 • Developed deep learning methods and tools for social science data curation with Melissa Dell
TEACHING	Harvard University , Cambridge, Massachusetts • <i>Causal Inference in Economics</i> (ECON 2148, Graduate): Spring 2025, 2026 • <i>Time Series</i> (STAT 242, Graduate): Fall 2024, 2025 • <i>Unleashing Novel Data at Scale</i> (ECON 2355, Graduate): Spring 2021
INVITED TALKS	• 2026: ESIF Economics and AI+ML Meeting (Cornell, June); AI and Economics Summer Conference (Chicago Booth, August) • 2022: ASSA Annual Meeting (Virtual, January)
SOFTWARE	• hdmht-discovery: https://github.com/jscarlson/hdmht-discovery • NetworkTargeting (with Davide Viviano): https://github.com/dviviano/NetworkTargeting • EffOCR (with Tom Bryan and Melissa Dell): https://pypi.org/project/efficient-ocr • dcr (with Trevor Incerti and P.M. Aronow): https://github.com/jscarlson/dcr, https://github.com/jscarlson/stata-dcr
ACADEMIC SERVICE	<i>Referee:</i> Journal of Political Economy
LAST UPDATED	July 27, 2026